5G/6G COMMUNICATION-ECA5604

EXPT-10

5G Network Slicing Environment Using MATLAB

Aim

To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Objectives

Objective – 1: Analyze the Number of Network Slices

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB.

Observe how different slices can be configured to support diverse service requirements and applications.

Objective – 2: Compare Slice Utilization (%)

Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

Objective – 3: Analyze End-to-End Latency (ms)

Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency

values to understand the Quality of Service (QoS) requirements of various 5G applications.

Software Required

• MATLAB

Objective – 1

Analyze the Number of Network Slices

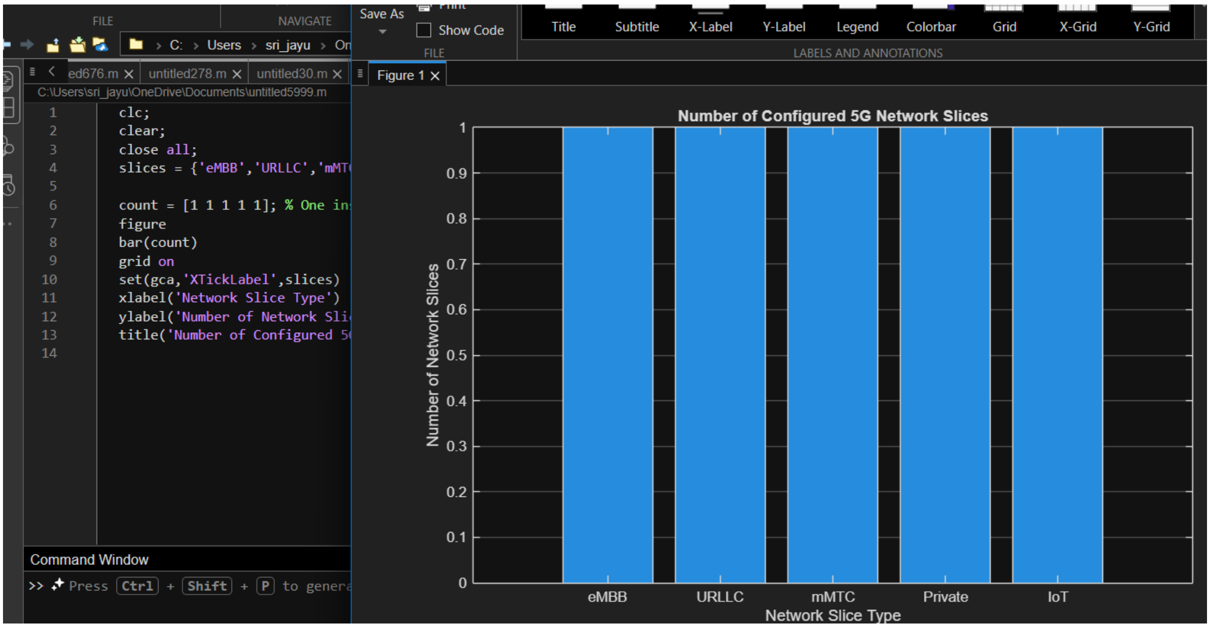
Generate and visualize multiple network slices using MATLAB. Observe how a 5G network can

be logically partitioned into independent slices for different communication services.

MATLAB Code:

```
clc;

clear;
```

```
close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};


count = [1 1 1 1 1]; % One instance of each network slice

figure

bar(count)

grid on

set(gca,'XTickLabel',slices)

xlabel('Network Slice Type')

ylabel('Number of Network Slices')

title('Number of Configured 5G Network Slices')
```



Objective – 2

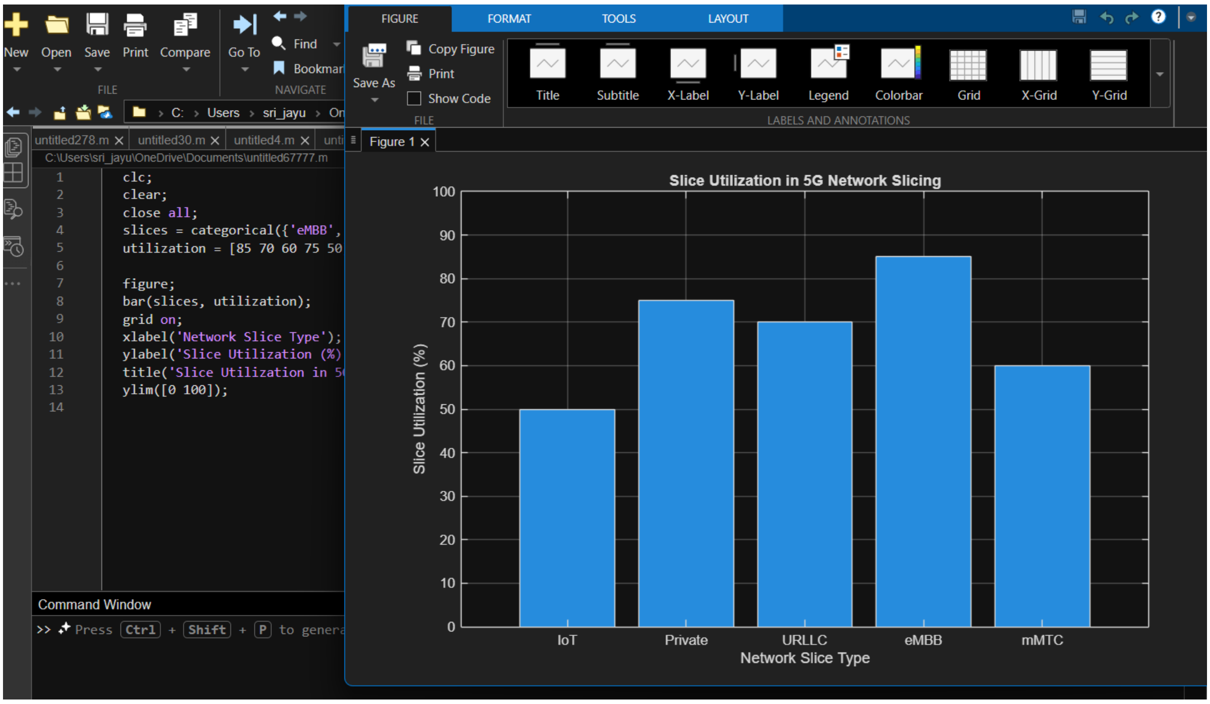
Compare Slice Utilization (%)

Analyze the utilization percentage of different network slices using MATLAB. Compare the resource usage of each slice to evaluate network efficiency.

MATLAB Code

```
clc;
```

```
clear;
close all;
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
utilization = [85 70 60 75 50];

figure;
bar(slices, utilization);
grid on;
xlabel('Network Slice Type');
ylabel('Slice Utilization (%)');
title('Slice Utilization in 5G Network Slicing');
ylim([0 100]);
```



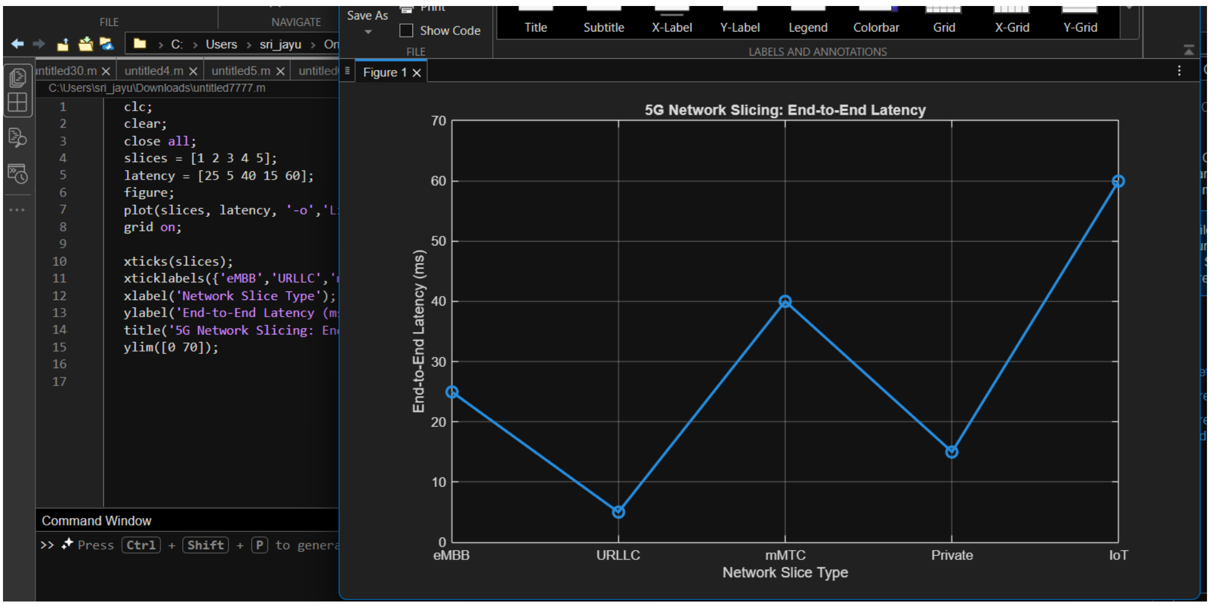
Objective – 3

Analyze End-to-End Latency (ms)

Compare the end-to-end latency of different network slices using MATLAB. Observe how latency varies depending on the service requirements of each slice.

MATLAB Code:

```
clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;

xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');
ylim([0 70]);
```



Result

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.